Michael Shepherd

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EDUCATION

The Pennsylvania State University | University Park, PA Anticipated Graduation: May 2027

B.S. in Applied Data Science, College of Information Sciences and Technology

TECHNICAL SKILLS

Programming Languages: Python | R | SQL | TypeScript | HTML/CSS
Libraries and Frameworks: Pandas | NumPy | scikit-learn | LightGBM | CatBoost | tidyverse | ggplot2 | FastAPI | Streamlit
Programs and Tools: PostgreSQL (pgvector) | SQLite | Docker | Git | GitHub | Vercel | Jupyter Notebook | Alteryx Designer | Microsoft Excel | Word | PowerPoint | Outlook | Autodesk Fusion | Ollama
Methods: Machine Learning | Regression | Classification | Clustering | Feature Engineering | Cross-Validation | RAG | REST APIs | ETL and Data Pipelines | Data Cleaning and Auditing | Data Visualization | Unit Testing
Spoken Language: Spanish (Fluent)

EXPERIENCE

Geoinformation and Big Data Research Laboratory (GIBD) University Park, PA

Undergraduate Research Assistant *May 2026 - Aug 2026*

- Built two Python agents for an existing geospatial research platform: one finds nearby NOAA flood alerts and the other prepares hazard records for mapping.
- Filtered alerts by location, distance and time. Standardized GeoJSON and CSV records, removed duplicates, rejected unsupported geometry and assigned four risk levels.
- Wrote 13 unit tests and validated the full workflow. A Houston demo combined 9 flood alerts and 3 earthquake records within 300 km.

Made By Design Lab University Park, PA

Undergraduate Research Assistant *August 2023 - May 2025*

- Developed and documented a repeatable multi-material 3D printing process using Mosaic hardware, Chroma software, iterative CAD designs and test prints.

American Society of Mechanical Engineers (ASME) Design Team University Park, PA

Chassis Subsystem Team Lead *January 2023 - May 2025*

- Led the chassis subsystem of a 3D-printed RC car. Designed drivetrain components and the gear differential casing in Autodesk Fusion and trained 8 team members.

PROJECTS

FutWAR: Soccer Wins Above Replacement Research (In Progress) | Python, PostgreSQL, SQL, Docker

- Researching player value across Europe's Big Five leagues from 2017-18 through 2024-25 using replacement-level baselines, league strength and positional roles.
- Built dataset audits, a local PostgreSQL workflow and event-context experiments. Published a player explorer prototype with data sources and documented limits.

Penn State DS 310 Kaggle Competitions | Python, scikit-learn, LightGBM, CatBoost, TabICL

- Regression: Team 9 placed first on the private leaderboard (MSE 2058.85). Compared ridge, PCA and SVR on 242 training rows before blending predictions.
- Classification: Team 9 placed third on the private leaderboard (ROC AUC 0.687). Blended tree ensemble and TabICL prediction ranks for the stock buy task.

NBA All-Star Prediction and Snub Exploration | Python, scikit-learn, Jupyter Notebook

- Modeled selections across approximately 3,800 NBA player seasons. Studied confident misses, including James Harden in 2023 (85.4% predicted, not selected).
- Tested True Shooting Percentage and team wins across five versions. Excluded them from the final analysis after data merges reduced the sample.

Spotify Playlist Curator | Python, Spotify Web API, OAuth 2.0 (PKCE), SQLite, Streamlit

- Built a seven-phase app that collects Spotify, MusicBrainz and Last.fm data, deduplicates recordings, tracks sources in SQLite and scores tracks with transparent rules.
- Added preview, approval, write verification and rollback for playlist updates. The project includes 265 automated tests.

Basketball RAG Assistant | Python, FastAPI, PostgreSQL (pgvector), Ollama, Docker

- Built a local chat app that retrieves relevant sample game records with pgvector before sending them to Llama. The interface displays the records behind each answer.

CERTIFICATES

IBM Data Analyst Professional Certificate | Alteryx Designer Core Certification